

The Erdős Discrepancy Problem

Sequence $c_n \in \{-1, +1\}$

+1, -1, -1, +1, +1, -1, +1, -1, -1, +1, +1, -1, +1, -1, -1, +1, ...

Partial Sums Along Arithmetic Progressions



The Question

Can we keep ALL partial sums bounded simultaneously?

NO! Erdős conjectured (1932): $\sup_n |\sum c_{dn}| = \infty$

PROVED by Tao (2015) using multiplicative functions + Chowla's conjecture